Practical 2: Data Preprocessing and Association Rule Mining using WEKA

Objective: Perform data preprocessing tasks and demonstrate association rule mining using Apriori and FP-Growth algorithms in WEKA.

Dataset Used:

```
@relation supermarket

@attribute Bread {yes,no}
@attribute Butter {yes,no}
@attribute Milk {yes,no}
@attribute Jam {yes,no}
@attribute Juice {yes,no}

@data
yes,yes,yes,no,no
yes,no,yes,yes,no
yes,yes,no,yes,no
no,yes,yes,no,yes
yes,yes,yes,yes,no
no,no,yes,yes,yes
yes,no,no,yes,yes
yes,yes,yes,no,yes
```

Step 1: Open WEKA Software

1. Click on Start Menu.
2. Search for WEKA.
3. Open **WEKA GUI Chooser**.
4. Click on **Explorer** option.
5. WEKA Explorer window will open.

Step 2: Create Dataset File

1. Open Notepad.
2. Copy the provided ARFF dataset.
3. Click File → Save As.
4. Enter file name as **supermarket.arff**.
5. Select Save as type → All Files.
6. Click Save.

Step 3: Load Dataset in WEKA

1. Go to **Preprocess** tab.
2. Click **Open File** button.
3. Browse the saved dataset file.
4. Select **supermarket.arff**.
5. Click Open.

6. Wait for dataset to load successfully.

Step 4: Verify Dataset Information

1. Observe relation name displayed on top.
2. Check number of instances.
3. Verify all attributes are loaded correctly.
4. Click each attribute to view:
• Attribute type
• Value distribution
• Histogram visualization
• Missing values information

Step 5: Perform Data Preprocessing

1. In Preprocess tab locate the **Filter** section.
2. Click **Choose** button.
3. Select:
unsupervised → attribute → Remove
4. Configure filter if needed.
5. Click **Apply** button.
6. Observe updated dataset.
7. Apply additional filters if required such as:
• ReplaceMissingValues
• Normalize
• Discretize

Step 6: Open Associate Tab

1. Click on **Associate** tab from top menu.
2. This tab is used for association rule mining.
3. WEKA supports algorithms such as:
• Apriori
• FP-Growth

Step 7: Select Apriori Algorithm

1. Click **Choose** button.
2. Navigate to:
associations → Apriori
3. Select Apriori algorithm.
4. Algorithm name will appear beside Choose button.

Step 8: Configure Apriori Parameters

1. Click on the algorithm name **Apriori**.
2. Configuration window will open.
3. Set parameters such as:
• Minimum support
• Minimum confidence

• Number of rules
• Metric type
4. Example configuration:
• minMetric = 0.9
• lowerBoundMinSupport = 0.1
• numRules = 10
5. Click OK.

Step 9: Execute Apriori Algorithm

1. Click **Start** button.
2. WEKA starts scanning dataset.
3. Frequent itemsets are generated.
4. Association rules are created.
5. Wait until processing completes.

Step 10: Observe Apriori Output

1. Output appears on right side panel.
2. Observe generated association rules.
3. Example rules:
• Bread=yes ==> Milk=yes
• Butter=yes ==> Bread=yes
4. Observe support and confidence values.
5. Analyze strongest association rules.

Step 11: Run FP-Growth Algorithm

1. Again click **Choose** button.
2. Select:
associations → FPGrowth
3. Click on FPGrowth to configure parameters.
4. Set minimum support value.
5. Click OK.

Step 12: Execute FP-Growth

1. Click **Start** button.
2. WEKA creates FP-Tree structure internally.
3. Frequent patterns are generated efficiently.
4. Wait for execution to complete.

Step 13: Analyze FP-Growth Output

1. Observe generated frequent itemsets.
2. Compare results with Apriori algorithm.
3. Observe support values.
4. Analyze frequent product combinations.
5. Identify highly associated items.

Step 14: Visualization and Analysis

1. Return to Preprocess tab.
2. Select attributes individually.
3. Observe histogram distributions.
4. Analyze attribute frequency visually.
5. Compare item occurrence patterns.

Step 15: Final Observation

1. Dataset preprocessing completed successfully.
2. Apriori generated association rules.
3. FP-Growth generated frequent patterns.
4. Product relationships identified successfully.
5. Data mining process completed successfully.

Step 16: Conclusion

Association rule mining was performed successfully using Apriori and FP-Growth algorithms in WEKA.
Data preprocessing operations were also executed successfully before mining the dataset.